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DISSERTATION

to obtain
the degree of doctor at the University of Twente,
on the authority of the rector magnificus,
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on account of the decision of the Doctorate Board,
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by

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That's all folks.

Abstract

Test

Samenvatting

Nederlands taal.

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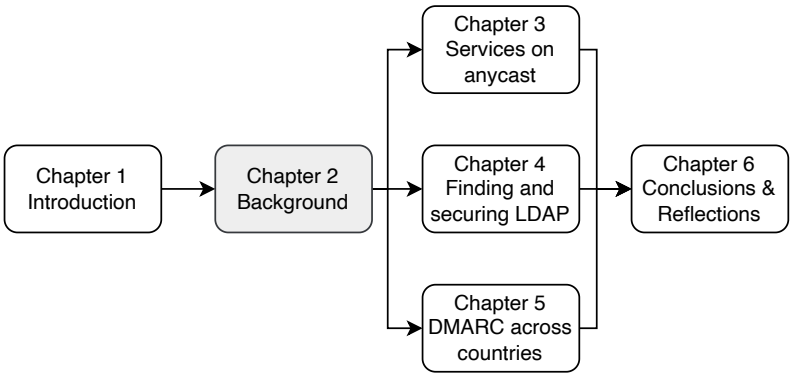
CHAPTER 1

Introduction

My intro [1].

CHAPTER 2

Background



This chapter provides background information on Anycast, Transport Layer Security (TLS), Lightweight Directory Access Protocol (LDAP), and email security protocols such as Sender Policy Framework (SPF), DomainKeys Identified Mail (DKIM), and Domain-based Message Authentication, Reporting and Conformance (DMARC). This information is necessary to to understand the subsequent chapters. Our goal is not to provide an exhaustive description of the aforemetioned topics, but rather offer sufficient details relevant to this thesis.

2.1 Anycast

The first RFC describing anycast (RFC 1546 [2]) was published in 1993 and explains how anycast simplifies finding an appropriate server. As service examples, they list DNS, FTP, and "Archie" (one of the first Internet search engines). Nowadays, anycast is widely deployed for critical Internet infrastructure, such as the DNS [3], or by Content Delivery Networks (CDNs) to provide web services [4, 5].

Anycast is the practice of making a service IP address (either IPv4 or IPv6) available in multiple physical locations [6]. Such locations are referred to catchments or simply as Points of Presence (PoPs). For instance, when a user is located in Enschede trying to access a website that is anycasted in multiple locations such as Amsterdam, Rio de Janeiro and Sydney, the user's request will be sent to Amsterdam. The routing decision is made by Border Gateway Protocol (BGP) which directs the request to a single PoP, typically one that is topologically close to the user [6] from a routing, and often geographically [7], perspective.

A common misconception is that announcing an address at multiple locations, in itself, is classified as anycast. Many CDNs announce their prefixes at multiple network edges (in different geographic regions), where they route traffic internally to a single PoP hosting the service. Critically, as defined in RFC 4786 [6], anycast is the replication of a service address where several PoPs make the service available.

There are multiple reasons for providing a service using anycast. Anycast can increase service resilience through automatic failover. For instance, in our previous example, if the Amsterdam PoP is unavailable, the user's request would be redirected to the next best PoP (either Rio de Janeiro or Sydney, depending on the routing decisions). Furthermore, anycast offers service replication closer to users to reduce latency, and enables load distribution across PoPs.

2.2 TLS

Secure communications was first developed by Netscape in 1994 with the introduction of Secure Socket Layer (SSL) version 1.0 [8]. SSL is a cryptographic protocol that Over time, SSL evolved where its last and deprecated version was 3.0. Internet Engineering Task Force (IETF) standardized the protocol as TLS in 1999 with version 1.0 [9]. The latest version of TLS is 1.3 [10] is the current recomended version, while 1.2 [11] is still widely used.

2.3 LDAP

2.4 Email Security Protocols

2.4.1 DKIM

2.4.2 SPF

2.4.3 DMARC

This chapter is either extended or adapted version of the author's publications. The Section 2.1 is an adapted version of [12].

CHAPTER 3

Sample Chapter

Sample chapter [1].

CHAPTER 4

Conclusions

Some conclusions.

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Publications by the Author

Below is a list of peer-reviewed academic publications that ... has authored or co-authored in reverse chronological order:

- usenix
- cnsm
- noms
- tma